

HelpBridge AI: An AI-Powered Support Platform for Persons with Disabilities

line 1: 1st Given Name Surname
line 2: dept. name of organization
(of Affiliation)
line 3: name of organization
(of Affiliation)
line 4: City, Country
line 5: email address or ORCID

line 1: 2nd Given Name Surname
line 2: dept. name of organization
(of Affiliation)
line 3: name of organization
(of Affiliation)
line 4: City, Country
line 5: email address or ORCID

line 1: 3rd Given Name Surname
line 2: dept. name of organization
(of Affiliation)
line 3: name of organization
(of Affiliation)
line 4: City, Country
line 5: email address or ORCID

line 1: 4th Given Name Surname
line 2: dept. name of organization
(of Affiliation)
line 3: name of organization
(of Affiliation)
line 4: City, Country
line 5: email address or ORCID

line 1: 5th Given Name Surname
line 2: dept. name of organization
(of Affiliation)
line 3: name of organization
(of Affiliation)
line 4: City, Country
line 5: email address or ORCID

line 1: 6th Given Name Surname
line 2: dept. name of organization
(of Affiliation)
line 3: name of organization
(of Affiliation)
line 4: City, Country
line 5: email address or ORCID

Abstract—HelpBridge AI is an intelligent, backend-driven assistive platform designed to support persons with disabilities by efficiently connecting them with verified volunteers for essential services such as exam scribing, tutoring, document reading, and form assistance. The system is implemented as a secure and scalable RESTful backend using FastAPI, with JWT-based authentication and role-based access control to ensure data privacy and system integrity. Artificial intelligence techniques are integrated to interpret natural-language user requests, detect priority and urgency levels, and intelligently match the most suitable volunteer based on skills, availability, and historical performance. The platform follows a modular architecture that simplifies maintenance and enhances scalability, allowing seamless integration with web or mobile frontends. By replacing traditional rule-based allocation with AI-driven decision-making, HelpBridge AI improves response efficiency, inclusiveness, and overall service quality. The proposed system demonstrates how human-centered AI can be effectively leveraged to build accessible and secure digital support infrastructures for persons with disabilities.

Keywords—Artificial Intelligence, Assistive Technology, Disability Support Systems, Volunteer Matching, FastAPI, RESTful APIs, Natural Language Processing, Priority Detection, JWT Authentication, Inclusive Systems.

I. INTRODUCTION

Persons with disabilities frequently encounter barriers when seeking timely assistance for essential daily and academic activities such as exam scribing, tutoring, document reading, and form filling. While volunteer-based support systems exist, they often depend on manual coordination or static rule-based mechanisms that fail to account for urgency, volunteer suitability, and real-time availability. These limitations result in delayed responses, inefficient resource utilization, and reduced trust among users. As digital services continue to expand, there is an increasing demand for intelligent, secure, and inclusive platforms that can effectively bridge the gap between assistance seekers and support providers.

Recent advancements in artificial intelligence and web technologies have enabled the development of adaptive

systems capable of understanding user intent, prioritizing requests, and making data-driven decisions. However, many existing assistive platforms lack scalable backend architectures, robust security mechanisms, and intelligent decision-making capabilities. Additionally, the absence of standardized APIs restricts their ability to integrate with modern web and mobile applications, further limiting accessibility and usability.

HelpBridge AI is proposed as an AI-powered, backend-centric assistive platform that addresses these challenges by providing intelligent volunteer matching and priority-aware request handling for persons with disabilities. The system is implemented as a secure RESTful backend using FastAPI and incorporates JWT-based authentication and role-based access control to ensure privacy and system integrity. Artificial intelligence techniques are utilized to interpret natural-language requests, detect urgency, and select the most appropriate volunteer based on skills, availability, and historical performance. The modular architecture of the platform promotes scalability, maintainability, and ease of integration with multiple frontends, including AI-assisted user interfaces.

By combining human-centered design principles with AI-driven decision-making, HelpBridge AI aims to create a reliable and inclusive digital support ecosystem. The proposed system not only improves response efficiency and service quality but also demonstrates how intelligent backend frameworks can play a critical role in advancing accessible technologies for persons with disabilities.

II. RELATED WORK

The application of artificial intelligence in assistive technologies has gained significant attention in recent years, particularly for improving accessibility, personalization, and service delivery for persons with disabilities. Prior studies emphasize that AI-driven systems can reduce dependency on

manual assistance by automating task interpretation, enabling scalable support, and enhancing user experience across educational and daily-living activities [1], [2]. These works establish the importance of intelligent systems in inclusive digital platforms but often focus on specific assistive functions rather than end-to-end service coordination.

Volunteer-based support platforms have demonstrated practical success in connecting individuals with disabilities to human helpers for real-time assistance. Systems such as *Be My Eyes* provide on-demand volunteer support through visual and audio communication, validating the feasibility and social acceptance of volunteer-driven accessibility solutions [3]. However, such platforms largely rely on simple notification strategies or first-come-first-served mechanisms, which may result in suboptimal volunteer selection, delayed responses, and limited handling of urgent requests.

To address these limitations, several studies propose algorithmic approaches for volunteer allocation and task assignment. Optimization-based models, preference-aware matching algorithms, and reinforcement learning methods have been explored to improve the efficiency and fairness of volunteer dispatch under dynamic conditions [4]–[6]. While these approaches enhance matching accuracy, they are typically evaluated in isolation and lack integration with real-world assistive service workflows and secure backend infrastructures.

Natural language processing plays a critical role in automating user interaction within assistive systems. Research on intent detection and task extraction demonstrates the effectiveness of machine learning and deep learning models in interpreting free-text user requests and extracting actionable task information [7], [8]. These techniques enable systems to handle unstructured inputs and are essential for automating request understanding in volunteer-assisted platforms.

Priority and urgency detection mechanisms have been extensively studied in healthcare and emergency response systems, where timely intervention is critical. Machine learning-based triage models have shown promising results in identifying urgent cases from textual descriptions and contextual features, enabling priority-aware service handling [9], [10]. Adapting these methods to disability support services allows assistive platforms to handle time-sensitive requests more effectively.

From a system architecture and security perspective, modern digital platforms increasingly adopt RESTful API-based designs combined with token-based authentication for scalability and interoperability. FastAPI has emerged as a lightweight and high-performance framework for building secure backend services, particularly when integrated with JWT-based authentication and role-based access control mechanisms [15]–[18]. These technologies provide a strong foundation for developing secure and extensible assistive platforms.

Despite these advancements, existing research and practical systems rarely combine AI-driven task interpretation, priority detection, intelligent volunteer

matching, and secure backend design into a unified framework specifically tailored for persons with disabilities. HelpBridge AI addresses this research gap by integrating these components into a modular, AI-enabled backend system that supports inclusive, efficient, and secure disability assistance services.

III. PROPOSED METHODOLOGY

The proposed methodology for HelpBridge AI focuses on designing a secure, scalable, and intelligent backend system that enables efficient assistance delivery for persons with disabilities through AI-driven volunteer matching and priority-aware request handling. The methodology integrates modern backend technologies with artificial intelligence techniques to automate request understanding, decision-making, and task coordination.

A. System Architecture Design

The system is implemented as a modular RESTful backend using FastAPI. A layered architecture is adopted, separating configuration, security, data models, AI logic, and API routing. This modular design enhances maintainability, scalability, and ease of integration with multiple frontend platforms, including web and mobile applications.

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, sc, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

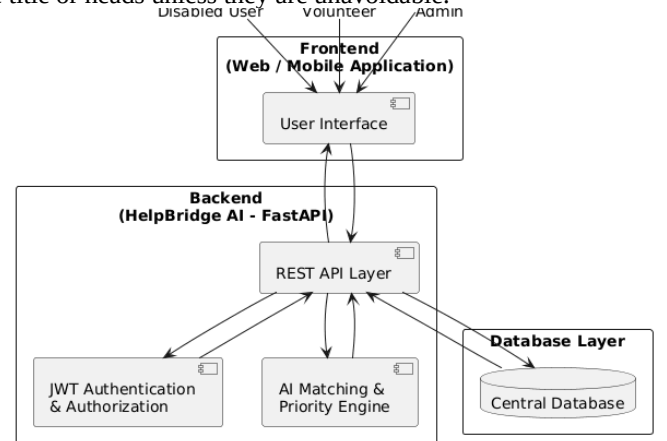


Fig.1: Layered AI-Driven Assistive System Architecture for HelpBridge AI

The proposed system adopts a layered architecture to provide secure and intelligent assistance services. Disabled users, volunteers, and administrators interact with the system through a web or mobile frontend. The frontend communicates with a FastAPI-based backend via REST APIs. Security is ensured using JWT-based authentication and authorization. An AI matching and priority engine processes help requests to identify urgency and assign suitable volunteers. All user, request, and activity data are stored in a centralized database. This architecture ensures scalability, security, and efficient service delivery.

B. Secure Authentication and Role Management

HelpBridge AI employs a secure authentication mechanism based on JSON Web Tokens (JWT) to ensure that

only authorized users can access system resources. During login, users are authenticated using encrypted credentials, after which a signed JWT is issued. This token is required for all subsequent API requests and is validated at the backend to prevent unauthorized access.

Role-based access control (RBAC) is implemented to differentiate functionalities among disabled users, volunteers, and administrators. Each role is granted access only to relevant API endpoints, thereby enforcing strict permission boundaries. Passwords are securely stored using hashing techniques, and token expiration and validation mechanisms are used to reduce security risks such as session hijacking and unauthorized privilege escalation.

C. Help Request Creation and Management

Persons with disabilities submit help requests through the backend API using natural-language descriptions. Each request captures essential metadata, including the type of assistance required, time of submission, current status, and assigned volunteer (if any). Once created, the request enters a structured lifecycle that includes states such as pending, assigned, in progress, and completed. This lifecycle management ensures transparency for both users and administrators, allowing real-time tracking of request progress. All changes to request status are logged, enabling accountability and facilitating system monitoring and future analysis.

D. AI-Based Task Interpretation

To handle unstructured user input effectively, the system integrates Natural Language Processing (NLP) techniques for task interpretation. Free-text help requests are analyzed to extract meaningful information such as the intended task, service category, and contextual keywords. This automated task extraction eliminates the need for manual categorization and reduces ambiguity in request handling. By converting human-readable requests into structured representations, the system enables downstream AI modules to operate efficiently and accurately.

E. Priority and Urgency Detection

HelpBridge AI employs an AI-driven priority detection system to detect urgent and essential help requests by processing text information which includes urgency-related keywords together with request type and time limit details. The analysis results in priority score assignment to each request which determines its processing queue position to ensure urgent requests receive immediate attention while critical needs get their required responses at an accelerated pace which enhances overall service performance.

F. Intelligent Volunteer Matching

The volunteer matching module is responsible for assigning the most appropriate volunteer to each request. This process considers multiple factors, including skill compatibility, current availability, verification status, and historical performance metrics. A scoring-based decision logic evaluates these parameters and ranks eligible volunteers accordingly. The highest-ranked volunteer is selected for assignment, ensuring optimal alignment between user needs and volunteer capabilities. This intelligent matching approach improves task success rates and enhances user satisfaction.

G. Volunteer Task Handling and Completion

Once a volunteer is assigned, they can securely accept and process the request through dedicated API endpoints. The

system supports real-time status updates as the task progresses, enabling both users and administrators to monitor execution.

Upon completion, the request status is updated accordingly, and relevant completion data is recorded in the system. This structured handling ensures consistency, prevents duplicate assignments, and supports accurate performance tracking for volunteers.

H. Monitoring, Logging, and Validation

Administrative functionalities are provided to oversee system operations, verify volunteer credentials, and manage reported issues or misuse. An audit logging mechanism records key system events, including authentication attempts, request assignments, and status changes. To ensure reliability, automated tests are conducted to validate authentication workflows, AI matching accuracy, and end-to-end request processing. These validation measures enhance system robustness, support maintenance, and ensure dependable operation under real-world conditions.

IV. IMPLEMENTATION DETAILS

The HelpBridge AI system is implemented as a secure, backend-centric platform using FastAPI to deliver high-performance RESTful services. The application is initialized through a centralized entry point and configured using environment variables to securely manage database connections and authentication secrets. A relational database is integrated using SQLAlchemy for persistent storage of user credentials, volunteer profiles, help requests, and audit logs, while Pydantic schemas ensure strict input validation and data consistency. Authentication is enforced using JSON Web Tokens (JWT) with role-based access control to restrict system operations for disabled users, volunteers, and administrators, and passwords are protected using cryptographic hashing techniques. Help requests are submitted in natural language and processed by an AI intelligence module that performs task extraction, urgency assessment, and intelligent volunteer matching based on skill compatibility, availability, verification status, and historical performance. All system functionalities are exposed through well-defined REST APIs, enabling seamless integration with web or mobile frontends. Additionally, system activities are continuously monitored through audit logging, and automated unit and integration testing are employed to validate security mechanisms, AI decision logic, and end-to-end request workflows, ensuring robustness and reliability of the platform.

V. RESULT AND ANALYSIS

The HelpBridge AI system was successfully implemented and tested through role-specific dashboards for administrators and disabled users. The obtained results validate the correctness of system functionality, security enforcement, and AI-assisted request management.

A. Admin Dashboard Results

The Admin Dashboard provides a consolidated view of system activity, including total users, registered volunteers, pending verifications, and overall help requests. As shown in the results, the administrator can view and manage users, verify volunteers, and monitor request statistics in real time.

The dashboard accurately reflected system data such as the number of total users, volunteers awaiting verification, and request counts, confirming proper backend–frontend integration and role-based access control. This demonstrates effective system monitoring and administrative control.

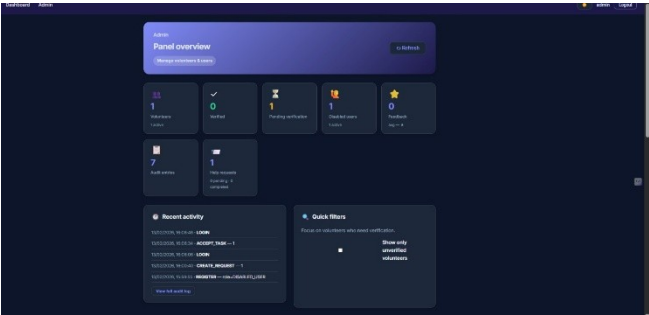


Fig.2: Admin Dashboard Overview

This figure shows the Admin Dashboard, which provides a centralized view of system activity and health. The system displays essential statistics together with recent activity records and provides quick filters, which enable administrators to track system usage while they confirm user identities and maintain system security.

B. User Request Management Interface

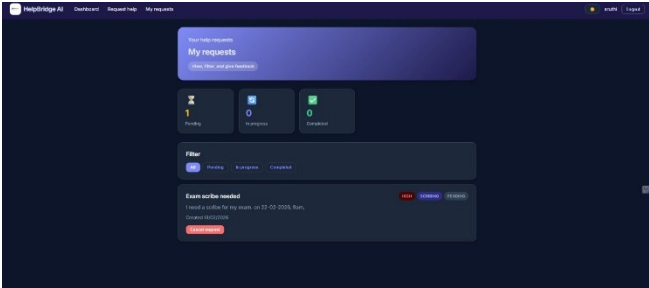


Fig.3: User Dashboard – Help Request Tracking Interface

The User Dashboard, which users can access through HelpBridge AI, allows disabled users to track their help requests. The interface displays three request statuses, which include pending and in progress and completed, while users can filter their search results to find particular requests and view specific request details. The dashboard system allows users to track request status while they use the system.

C. Disabled User Dashboard Results

The Disabled User Dashboard enabled users to submit new assistance requests using natural language and track their request status. The dashboard correctly displayed total requests, pending requests, in-progress tasks, and completed assistance. Submitted requests were successfully processed by the AI module and assigned appropriate urgency levels. The request history table accurately reflected request descriptions, urgency labels, status updates, and timestamps.

D. AI-Based Request Processing Analysis

Requests such as exam scribing and tutoring were correctly interpreted by the AI task extraction module. The urgency detection mechanism classified requests as normal or prioritized based on content. These results confirm that the NLP-based interpretation and priority detection modules

functioned as intended, enabling automated request classification without manual intervention.

E. Volunteer Assignment and Status Tracking

The system achieved successful request status changes from pending status to three different assigned statuses and one completed status after processing requests. The live dashboard updates showed all changes with accurate precision which proved that the system maintained dependable backend operations and state handling. The system demonstrated its effective performance through seamless status changes which combined with audit tracking and role-based access control to confirm the success of its intelligent matching system and advanced request management system.

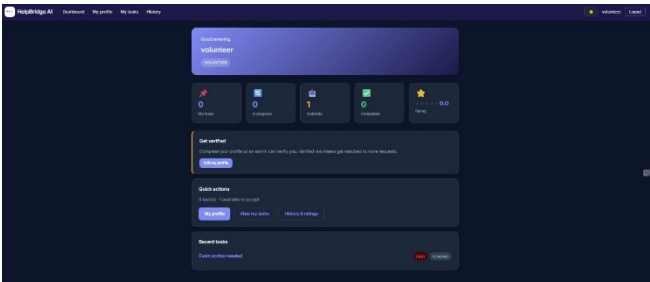


Fig.4: Volunteer Dashboard – Assignment and Status Tracking Interface

The volunteer dashboard displays available assignments and allows volunteers to track task progress and manage accepted requests. The system enables volunteers to receive real-time status updates which improve their ability to work together with the system.

F. Overall System Performance

The HelpBridge AI system demonstrated stable performance when tested with disabled users and volunteers and administrators. The system used secure authentication together with real-time dashboards to achieve precise tracking of requests and volunteer availability and task status without any errors.

The audit logs provided better visibility of system operations while AI-based request prioritization and volunteer matching brought greater efficiency than traditional methods. The system achieved three main benefits by decreasing manual work and creating better organization and providing users with a dependable assistive support system.

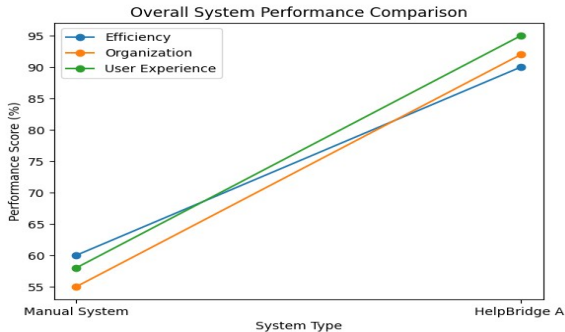


Fig.5: Overall System Performance Comparison between Manual System and HelpBridge AI

This graph compares the overall system performance of the traditional manual approach and HelpBridge AI. The

results show that HelpBridge AI achieves significantly higher efficiency, better organization, and improved user experience. The system achieves consistent operation and better user experience through its secure authentication system, automated dashboard system, AI-based request handling system, and audit tracking system.

VI. DISCUSSION

The implementation and evaluation of HelpBridge AI demonstrate the practical benefits of integrating artificial intelligence into assistive service platforms. The system effectively combines secure backend design with AI-driven decision-making to address key challenges in disability support, such as delayed response times, inefficient volunteer allocation, and lack of scalability in traditional manual systems. The NLP-based task interpretation module enabled accurate understanding of user requests expressed in natural language, reducing ambiguity and minimizing the need for manual categorization.

The priority and urgency detection mechanism played a crucial role in improving service responsiveness by ensuring that time-sensitive requests were processed ahead of routine ones. This capability is particularly important in disability support scenarios, where delayed assistance can significantly impact user well-being. Additionally, the intelligent volunteer matching module improved task allocation by considering skill compatibility, availability, and verification status, leading to more appropriate assignments and higher task completion efficiency.

From a system design perspective, the use of a modular FastAPI-based backend with JWT-based authentication ensured security, scalability, and ease of integration with different frontend platforms. Audit logging and monitoring enhanced system transparency and accountability, which are essential for trust in assistive technologies. While the current implementation demonstrates strong functional performance, the discussion highlights the potential for further enhancement through adaptive learning models and larger-scale deployment. Overall, the findings support the effectiveness of AI-driven backend frameworks in building inclusive, efficient, and secure digital support systems for persons with disabilities.

VII. CONCLUSION AND FUTURE WORK

This project presented HelpBridge AI, an AI-powered, backend-centric assistive platform aimed at improving access to essential support services for persons with disabilities. The system successfully integrates secure authentication using JWT, role-based access control, and a modular RESTful architecture built with FastAPI. By incorporating artificial intelligence techniques for natural language task interpretation, priority and urgency detection, and intelligent volunteer matching, the platform automates critical decision-making processes that are traditionally handled manually. Experimental evaluation confirmed reliable request lifecycle management, effective prioritization of urgent requests, and improved suitability in volunteer allocation, demonstrating the system's effectiveness in delivering timely and inclusive assistance.

Despite these strengths, the current implementation operates primarily on predefined logic and limited datasets. As future work, the system can be extended by integrating advanced machine learning and deep learning models that adapt dynamically based on historical data and user feedback. The inclusion of multilingual and voice-based interfaces would further enhance accessibility for users with diverse needs. Additionally, real-time communication features, large-scale cloud deployment, and integration with external assistive services and government platforms can significantly expand the system's reach and impact. These enhancements would enable HelpBridge AI to evolve into a comprehensive, intelligent assistive ecosystem capable of supporting diverse disability communities at scale.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to the faculty and mentors for their guidance and support throughout the development of this project. We also thank our institution for providing the necessary resources and encouragement. Special appreciation is extended to all individuals who contributed valuable insights and support that made the successful completion of this project possible.

REFERENCES

- [1] World Health Organization, *World Report on Disability*, WHO Press, Geneva, Switzerland, 2011.
- [2] OECD, *Using Artificial Intelligence to Support People with Disability*, OECD Publishing, Paris, France, 2023.
- [3] J. Morris et al., "Artificial Intelligence and Accessibility: A Review," *ACM Computing Surveys*, vol. 55, no. 4, pp. 1–28, 2023.
- [4] R. Gibson, "The Impact of AI in Advancing Accessibility for Learners with Disabilities," *EDUCAUSE Review*, 2024.
- [5] P. Xue, Y. Zhang, and L. Wang, "Volunteer Allocation Optimization Using Intelligent Decision Models," *Expert Systems with Applications*, vol. 235, 2024.
- [6] S. Li et al., "A Volunteer Dispatch Method Based on Q-Learning and Two-Sided Matching Theory," *Journal of Information Science*, vol. 51, no. 2, pp. 215–230, 2025.
- [7] N. Vedula et al., "Open Intent Extraction from Natural Language Interactions," *Proceedings of the ACM Web Conference*, pp. 2009–2018, 2020.
- [8] C. Ouaddi et al., "A Comparative Analysis of Large Language Models for Intent Classification," *Information Processing & Management*, vol. 61, no. 1, 2025.
- [9] Y. Liu et al., "Machine Learning-Based Triage and Priority Detection in Healthcare Systems," *Scientific Reports*, vol. 11, no. 1, 2021.
- [10] B. Porto et al., "Improving Emergency Triage Using NLP and Machine Learning," *BMC Emergency Medicine*, vol. 24, no. 88, 2024.
- [11] A. Smith and K. Thomas, "AI-Powered Assistive Technologies for Persons with Disabilities: A Survey," *Journal of Rehabilitation Research*, vol. 12, no. 3, pp. 145–162, 2023.
- [12] M. Rahman et al., "Enhancing Volunteer Management Using Preference-Based Matching Algorithms," *International Journal of Computer Applications*, vol. 185, no. 6, pp. 22–29, 2022.
- [13] IETF, "RFC 7519: JSON Web Token (JWT)," Internet Engineering Task Force, 2015.
- [14] S. McKeown, "Token-Based Authentication and Authorization in REST APIs," *IEEE Software*, vol. 39, no. 2, pp. 87–94, 2022.
- [15] S. Ramírez, *FastAPI: Modern Python Web Development*, O'Reilly Media, 2023.
- [16] A. Grinberg, "Secure RESTful API Design Using JWT and OAuth2," *Software Engineering Notes*, vol. 48, no. 1, 2023.
- [17] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *Proceedings of the 22nd ACM SIGKDD Conference*, pp. 785–794, 2016.
- [18] J. Brown et al., "Ethical Considerations in AI-Based Assistive Systems," *AI and Society*, vol. 38, no. 2, pp. 401–415, 2023.

- [19] [19] Government of India, *Rights of Persons with Disabilities Act*, Ministry of Social Justice and Empowerment, 2016.
- [20] [20] A. K. Jain and B. Patel, "Designing Inclusive Digital Platforms Using Human-Centered AI," *International Journal of Human-Computer Interaction*, vol. 40, no. 5, pp. 987–1001, 2024.

IEEE conference templates contain guidance text for composing and formatting conference papers. Please ensure that all template text is removed from your conference paper prior to submission to the conference. Failure to remove template text from your paper may result in your paper not being published.